Polynomial-time parity-query policies for explicit Boolean polynomials

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Abstract

Given the explicit squarefree algebraic normal form of a Boolean polynomial p of degree d, we construct an on-demand randomized parity-query policy with polynomial total bit time and working space. For every fixed hidden input, with probability at least $1-\delta$, it evaluates p correctly using $\min\{n, O(1+4^d \log s)\}$ queries, where s is the Fourier sparsity of $(-1)^p$. Polynomial anchors in one variable and a bounded kernel-basis repair remove the extra logarithmic query factor from the previous version. The polynomial runtime exponents are independent of d; time, space, termination and the cap of n queries hold on every random tape. A factor-partition recurrence supplies canonical restricted forms from the original source and a compact affine map, and a spectral-norm cutoff pays its exponential tables. The construction relies on attributed structural results of Zhang and Moshkovitz–Zhu, with Fourier data used only in the analysis. We retain a separate all-coset one-batch theorem and provide a complete implementation with bounded exact validation, including total and partial rank loss at specialization. The guarantee is randomized and per input; it does not construct a globally correct sampled deterministic tree or improve the known existential depth bound.

1 Statement and relationship to prior work

All logarithms are base two unless written ln. For $f : \mathbb{F}_2^n \to \{-1, 1\}$ let

$$\widehat{f}(\xi) = 2^{-n} \sum_x f(x)(-1)^{\xi \cdot x}, \qquad B = \|\widehat{f}\|_1, \qquad s = |\operatorname{supp} \widehat{f}|.$$

A parity query asks for a binary linear form in the unknown input. Independent equations for a subspace of codimension r specify r such queries; their answers identify a coset. Degree throughout means the degree of the unique squarefree algebraic normal form (ANF) over $\mathbb{F}_2$, not real Fourier degree.

The degree-reduction approach to parity decision trees predates this note [3]. Zhang [1] proves an existence bound of order $1 + 4^d \log s$, using a generic contraction radical and dimension-preserving descent. Remark 3.17 of that work expressly leaves construction time open. Earlier versions used a directly computable binary intersection with a logarithmic query loss. Here a bounded one-variable implementation of descent removes that loss while retaining polynomial computation. This addresses a randomized visited-path contract, not the construction of an everywhere-correct deterministic tree. The generic-rank comparison used below is due to Moshkovitz and Zhu [2]. The main contribution here is a uniformly polynomial-time complete visited-path construction. It combines canonical component evaluation, rank-preserving polynomial sampling, valuation-budgeted basis repair and a query-preserving cutoff. The descent principle is attributed to Zhang; the explicit degree and repair budgets are essential to the computational guarantee. A separately useful one-batch theorem has an all-coset guarantee. Both rely on the attributed structural interface; no priority claim for the combination is made.

For recursion, separate the original polynomial's encoding length S_0 from the accuracy input. An integer $b \geq 2$ requests failure at most $\epsilon = 2^{-b}$. For any desired $0 < \delta < 1$, choose $b = \max\{2, \lceil \log(1/\delta) \rceil\}$, so $\epsilon \leq \delta$. A binary-fraction prefix through the first nonzero bit also suffices. An arbitrary numerator/denominator encoding of length A, if used instead, adds its own $\mathrm{poly}(A)$ parsing cost; it cannot be charged solely to $\log(1/\delta)$.

Theorem 1. *Given the original explicit ANF of degree $3 \leq d \leq n$, precision $b \geq 2$, and parity-query access to a hidden $x \in \mathbb{F}_2^n$, there is a randomized on-demand policy with the following properties. Set*

$$k = \lceil \log(8nd) \rceil, \qquad t_A = b + \lceil \log(2d) \rceil.$$

For every fixed x, with probability at least $1 - 2^{-b}$, the policy returns $p(x)$ correctly and uses at most

$$\min\left\{n,\ 1 + \frac{2^{2d-1} - 5}{3} \log B\right\} \leq \min\left\{n,\ 1 + \frac{2^{2d-1} - 5}{6} \log s\right\} \tag{1}$$

queries, hence $\min\{n, O(1 + 4^d \log s)\}$. On every random tape it terminates, asks at most n queries, uses polynomial working space, and has bit time

$$\mathrm{poly}(n, S_0, b),$$

with a polynomial exponent independent of d. It uses neither Fourier information nor a residual-degree oracle and prints only the visited path. Original degree at most two is handled exactly in polynomial time by the leaf routine below.

Theorem 1 removes version 0.3.0's factor k multiplying $(2^{2d-1} - 8)/3$ in the query coefficient, retaining uniform polynomial computation and the same error contract. The degree dependence in the number of queries remains exponential. We also retain the following all-coset batch result.

Theorem 2. *Let $p \in \mathbb{F}_2[x_1, \ldots, x_n]$ be given by its L distinct squarefree nonzero monomials, of actual degree $3 \leq d \leq n$. Let $0 < \delta \leq 1/4$ be rational and let S be the encoding length, including δ. Put*

$$k = \left\lceil \log \frac{4n(d-2)}{\delta} \right\rceil.$$

There is a randomized algorithm whose worst-case bit cost is polynomial in n, S and $\log(1/\delta)$. It returns a basis matrix C for a binary subspace U and independent equations $Hx = 0$ for U, or declares failure. With probability at least $1 - \delta$ it returns matrices satisfying, simultaneously for every $a \in \mathbb{F}_2^n$,

$$\deg_{\mathbb{F}_2}(p(a + Cz)) \leq d - 1, \tag{2}$$

$$r := \mathrm{rank}_{\mathbb{F}_2} H = \mathrm{codim}\, U \leq \min\{n, k\, 2^{2d-3} \log B\} \leq \min\{n, k\, 2^{2d-4} \log s\}. \tag{3}$$

The degree guarantee is exact on success. Both declared failure and incorrect output count against δ. Space is polynomial and the output length is $O(n^2)$ bits. The algorithm needs no Fourier data, B, s, or generic-rank oracle.

The input model is essential. Polynomial time includes polynomial dependence on the ambient dimension n; S alone need not describe all unused variables. This is an explicit ANF input, not an arbitrary succinct arithmetic or Boolean circuit. The codimension estimate remains exponentially dependent on d, even though the construction time does not.

The batch guarantee is simultaneous over all cosets on its success event. The recursive guarantee concerns each fixed hidden input; it does not assert that one random seed defines a correct deterministic tree for all inputs.

2 The imported structural interface

Define $\Delta_h p(x) = p(x+h) + p(x)$ and

$$T(h_1, \dots, h_d) = \Delta_{h_1} \cdots \Delta_{h_d} p(0).$$

Formal evaluation uses the supplied squarefree polynomial over any extension of $\mathbb{F}_2$. The form T is symmetric, multilinear and alternating. Let $h = d-2$, let $y_1, \dots, y_h$ be independent formal n-vectors, and put

$$K = \mathbb{F}_2(y_1, \dots, y_h), \quad N(y)_{ij} = T(y_1, \dots, y_h, e_i, e_j), \quad R = \operatorname{rank}_K N, \quad W = \ker_K N.$$

The needed external inputs are

$$T(K^n, W, \dots, W) = 0, \qquad R \le 2^{d-2} M, \tag{4}$$

where the identity has $d-1$ copies of W, and M is the maximum rank of N at binary anchors. The first statement is Zhang's Theorem 3.12 [1]. The second is the generic/max-rank inequality of Moshkovitz–Zhu [2], applied to $h = d-2$ homogeneous multilinear blocks over $\mathbb{F}_2$. Its factor is 2^h and it has no large-field restriction. We import these structural statements rather than present independent proofs of them.

For completeness, the remaining quantitative connection is elementary. At binary anchors, $g = \Delta^{\times}_{y_1} \cdots \Delta^{\times}_{y_h} f$ is a quadratic phase, where $\Delta^{\times}_u f(x) = f(x+u)f(x)$. Its polar matrix is $N(y)$. If the polar rank is ρ, quadratic normal form gives $\|\widehat{g}\|_1 = 2^{\rho/2}$: each independent quadratic pair contributes a factor of two and affine terms only translate the spectrum. Translation preserves spectral norm and multiplication is Fourier convolution. Hence

$$2^{\rho/2} \le B^{2^h}, \qquad M \le 2^{d-1} \log B, \qquad R \le 2^{2d-3} \log B.$$

Parseval and Cauchy–Schwarz give $B \le \sqrt{s}$. These inequalities are used in the analysis; their parameters are never computed by the algorithm.

3 Specialization and binary equations

Lemma 3. *Let $N(y)$ be a polynomial matrix over $\mathbb{F}_2$ of generic rank R. Suppose its generic kernel W satisfies the first identity in (4) for a fixed multilinear form T over $\mathbb{F}_2$. For any field extension E and any specialization a with $\operatorname{rank}_E N(a) = R$, the kernel $W_a = \ker_E N(a)$ satisfies $T(E^n, W_a, \dots, W_a) = 0$.*

Proof. If $R = 0$, the matrix is identically zero and the assumed identity gives the claim. Otherwise choose R rows and columns with submatrix P such that $D = \det P$ is nonzero at a. Reorder columns and write those rows as $[P\ Q]$. Over the localization $\mathbb{F}_2[y, 1/D]$, the columns of

$$V = \begin{bmatrix} -P^{-1}Q \\ I_{n-R} \end{bmatrix}$$

are a basis of the generic kernel. They solve the selected equations, whose rows span the generic row space, so $NV = 0$. Every entry has denominator dividing D. For standard basis vectors e_i and columns of V, multiply each identity $T(e_i, V_{j_1}, \dots, V_{j_{d-1}}) = 0$ by D^{d-1}. It becomes a polynomial identity, which can be specialized at a. Since $D(a) \neq 0$, the original identity also holds there. The $n - R$ independent columns of $V(a)$ solve $N(a)V(a) = 0$ and span its entire kernel. Multilinearity proves the result. If $R = n$, the kernel has dimension zero and the assertion is immediate. □

This establishes preservation at every full-generic-rank point. No further denominator condition or union over pivot charts is needed. For probability analysis only, fix any nonzero generic R-minor D of N (use $D = 1$ if $R = 0$). Its degree is at most $R(d-2) \le n(d-2)$. For independent uniform anchors in $E = \mathbb{F}_{2^k}$, the polynomial zero bound gives

$$\Pr[\operatorname{rank} N(a) < R] \le \Pr[D(a) = 0] \le \frac{n(d-2)}{2^k} \le \delta/4. \tag{5}$$

The zero bound is the usual induction from the univariate root bound. The algorithm does not find D, determine R, or test this event. All larger minors vanish identically, so no specialization has rank greater than R.

For $A = N(a)$, select a row basis over E, of size r_E. Expand its entries in a fixed binary basis of E. For binary vectors x, each E-linear equation is exactly equivalent to at most k binary equations. Binary elimination produces H and C with

$$U = \ker_{\mathbb{F}_2} H = \ker_E A \cap \mathbb{F}_2^n, \qquad \operatorname{codim} U \le \min\{n, kr_E\} \le \min\{n, kR\}.$$

Thus (3) holds on every nonfailure output, even on exceptional rank samples. On the full-rank event Lemma 3 gives $T(U^d) = 0$. To obtain (2), each degree-d coefficient of the Boolean restriction $p(a + Cz)$ is a d-fold difference in the corresponding distinct columns of C. It equals T on those columns, independently of a, and therefore vanishes. The restriction has degree at most d before this argument. If $\dim U < d$, no such coefficient exists.

4 Polynomial evaluation and a capped algorithm

Lemma 4. *If $p = \sum_I \prod_{i \in I} x_i$ is squarefree of degree d, then over any field of characteristic two,*

$$T(h_1, \dots, h_d) = \sum_{|I|=d} \det\big[(h_j)_{i_\ell}\big]_{j,\ell=1}^{d}, \qquad I = \{i_1, \dots, i_d\}. \tag{6}$$

Consequently one value of T can be computed in $O(Ld^3)$ field operations.

Proof. Expand the subset formula for the d differences of one monomial. Every term omitting a direction occurs an even number of times and cancels. A monomial of degree less than d contributes nothing. For a degree-d monomial, each surviving term uses every direction exactly once, so terms are indexed by permutations. This is the permanent of the displayed matrix, equal to its determinant in characteristic two. Summing over monomials and using Gaussian elimination proves the cost. □

Only the listed degree-d monomials are needed. In particular, neither 2^d subset enumeration nor construction of an n^d tensor is necessary. Let

$$b = \lceil \log(2/\delta) \rceil, \qquad t = 2kb.$$

The following algorithm has fixed caps; all arithmetic is exact.

1. Read and check the explicit squarefree ANF and its actual degree. Compute k, b, t using integer comparisons with the rational input.

2. Try at most t independently uniform monic degree-k binary polynomials g, using k fair bits per trial. Test irreducibility by checking $\gcd(g, X^{2^j} - X) = 1$ for $1 \le j \le \lfloor k/2 \rfloor$, using repeated squaring modulo g. Keep the first passing polynomial; if none passes, declare failure.

3. Work in $E = \mathbb{F}_2[X]/(g)$ with its polynomial basis. Draw $d-2$ uniform independent anchors in E^n, using fresh random bits.

4. Form $A_{ij} = T(a_1, \ldots, a_{d-2}, e_i, e_j)$ by (6). Compute a row basis over E, expand it into binary equations, and compute H and C by binary elimination. Return these matrices without retrying on the basis of sampled rank.

The field test is exact: a reducible degree-k polynomial has an irreducible factor of degree $j \le k/2$, which divides $X^{2^j} - X$. An irreducible degree-k polynomial has none. Our parameters ensure $k \ge 6$. At least a $1/(2k)$ fraction of monic degree-k binary polynomials are irreducible. To see this, in $\mathbb{F}_{2^k}$ the elements of smaller degree lie in proper subfields. Their number is at most $\sum_{j=1}^{\lfloor k/2 \rfloor} 2^j \le 2^{k-1}$. Each irreducible degree-k polynomial accounts for k of the remaining elements. This is a counting argument, not field enumeration. Failure of all t trials has probability at most $\exp(-t/(2k)) = \exp(-b) \le 2^{-b} \le \delta/2$. Conditional on any verified field, fresh anchors satisfy (5). The total bad probability is at most $3\delta/4 \le \delta$.

Proposition 5. *Besides polynomial input-reading and validation overhead, the algorithm uses at most*

$$O\Big(tk^4 + (n^2Ld^3 + n^3)k^3 + kn^3\Big) \tag{7}$$

arithmetic bit operations with indexed storage. It uses at most $tk + (d-2)nk$ random bits and polynomial working space. Ordinary sequential access can also be charged while preserving a polynomial bit-time bound.

Proof. Schoolbook repeated squaring and polynomial gcd give the deliberately loose $O(k^4)$ bound per field trial. Field addition and multiplication cost at most $O(k^2)$ bits; inversion by exponentiation or polynomial Euclid fits in $O(k^3)$. Lemma 4 computes n^2 entries in $O(n^2Ld^3)$ field operations. Row elimination costs $O(n^3)$ such operations. Expanding rows gives at most kn binary equations of length n; elimination and a nullspace basis cost $O(kn^3)$ bit operations. The trial and anchor bit counts are explicit. Store one trial polynomial, all anchors, the two elimination matrices and one determinant workspace. A conservative space bound is $O(S + n^2k + ndk + d^2k + k^2 + b)$ bits. Address decoding or a sequential scan of these polynomial-size arrays adds only polynomial overhead. Since $d \le n$, L is bounded by the explicit description length and $k, b = O(\log(nd/\delta))$, these are polynomial costs in the parameters stated in Theorem 2. □

This proves Theorem 2. It supplies an algorithm and resource proof, not an implementation or performance benchmark.

5 Canonical residual forms

Maintain $a \in \mathbb{F}_2^n$ and a full-column-rank matrix $C \in \mathbb{F}_2^{n \times m}$ such that the possible hidden inputs are exactly $a + C\mathbb{F}_2^m$. The current Boolean function is $u(z) = p(a + Cz)$. Its unique squarefree representative is used in the proof but never expanded or stored. Evaluating p at a fully specified Boolean point is charged computation from its explicit description, not a query to the hidden x.

Lemma 6. *Suppose the Boolean degree of u is at most r. For binary directions define*

$$F_r(v_1, \ldots, v_r) = \sum_{J \subseteq [r]} p\left(a + C\sum_{j \in J} v_j\right) \quad \text{in } \mathbb{F}_2. \tag{8}$$

This is the canonical top r-form: it is independent of the base point, symmetric, alternating and $\mathbb{F}_2$-multilinear. It is zero if $\deg u < r$. *For a binary basis $\beta_0, \dots, \beta_{k-1}$ of E, decompose $h_i = \sum_j \beta_j v_{i,j}$. Its scalar extension is exactly*

$$T_E(h_1, \dots, h_r) = \sum_{j_1,\dots,j_r=0}^{k-1} \left(\prod_{i=1}^{r} \beta_{j_i} \right) F_r(v_{1,j_1}, \dots, v_{r,j_r}). \tag{9}$$

For $T_E(h_1, \dots, h_{r-2}, e_i, e_j)$, at most $k^{r-2}2^r$ original-source evaluations suffice. They can be streamed in polynomial space.

Proof. An $(r+1)$-fold difference of a Boolean polynomial of degree at most r vanishes. Thus the r-fold difference is independent of its base point. The operator identity $\Delta_{v+w} = \Delta_v + \Delta_w + \Delta_v\Delta_w$ proves additivity in every direction: its last term contributes an $(r+1)$-fold difference. Also $\Delta_v^2 = 0$, proving alternation; commutation gives symmetry. Values on distinct basis vectors extract the top coefficients of the canonical ANF. Formula (9) is now multilinearity, and in the contraction only the first $r-2$ slots need expansion. Nested tuple/subset loops accumulate one field value without storing their exponentially many states. □

This order of operations matters. For $p = x_1x_2 + x_1$ and $(x_1, x_2) = (z, z)$, the Boolean restriction and every value of (8) are zero. The unreduced formal expression $z^2 + z$ is nonzero at $\alpha \in E \setminus \mathbb{F}_2$. More sharply, restrict $x_1x_2x_3x_4$ to (z_1, z_1, z_2, z_3). Its canonical cubic form at $(\alpha e_1, e_2, e_3)$ equals α, whereas third field differences of the unreduced expression return α^2. Lemma 6 handles both controls in every dimension; it never evaluates the substituted formal expression at a nonbinary point.

An upper bound on the current degree suffices. The preceding evaluator is retained as an elementary baseline, but its degree-dependent cost does not establish uniform polynomial recursion. We next replace it with a canonical-component evaluator and then bound that evaluator's cost by a query-preserving cutoff.

5.1 Grouping Boolean collisions before exterior evaluation

Let E be a commutative ring of characteristic two and $r \geq 1$. Write

$$\mathcal{A}_r = E[e_1, \dots, e_r]/(e_1^2, \dots, e_r^2), \qquad w_j = \sum_{i=1}^{r} h_{ij} e_i$$

for given directions $h_1, \dots, h_r \in E^m$. This graded algebra is commutative, with basis e_I for $I \subseteq [r]$. Each $w_j^2 = 0$: cross terms cancel in characteristic two. It is not the Boolean quotient in which $z_j^2 = z_j$.

For one original monomial, let $D \leq d$ denote its number of factors, indexed by $\ell \in [D]$. After the affine substitution these factors are $a_\ell + \sum_j C_{\ell j} z_j$, where the subscripts select the corresponding original rows. Define

$$v_B = \sum_{j=1}^{m} \left(\prod_{\ell \in B} C_{\ell j} \right) w_j \qquad (\varnothing \neq B \subseteq [D]), \tag{10}$$

$$F_\varnothing = 1, \qquad F_A = \sum_{\substack{\varnothing \neq B \subseteq A \\ \min A \in B}} v_B F_{A \setminus B}, \qquad (\varnothing \neq A \subseteq [D]), \tag{11}$$

$$G = \sum_{A \subseteq [D]} \left(\prod_{\ell \notin A} a_\ell \right) F_A. \tag{12}$$

Lemma 7. *If u_M is the canonical squarefree Boolean representative of this restricted monomial, then $G = u_M(w_1, \dots, w_m)$. Summing over the original monomials yields $u(w)$. Its coefficient of $e_{[r]}$ equals*

$$\sum_{\substack{J\subseteq[m]\\|J|=r}} c_J \det(h_{ij})_{i\in[r],j\in J}, \qquad u(z) = \sum_J c_J \prod_{j\in J} z_j. \tag{13}$$

Under $\deg u \le r$ this is the required scalar-extended top form. No degree promise is needed for the component identity itself.

Proof. Induction shows that F_A is the sum over unordered set partitions π of A of $\prod_{B\in\pi} v_B$. Every partition has one unique block containing $\min A$, so it occurs exactly once. No factorial or division is used.

Expand this product by assigning a residual coordinate to each block. If two blocks receive the same coordinate j, the term contains $w_j^2 = 0$. The surviving assignments are injective on blocks. They correspond bijectively to maps $\phi : A \to [m]$ together with their partitions into nonempty fibers. Their coefficient is $\prod_{\ell\in A} C_{\ell,\phi(\ell)}$, and their value is $\prod_{j\in\operatorname{im}\phi} w_j$. In the original product expansion, factors outside A contribute constants and Boolean reduction replaces $\prod_{\ell\in A} z_{\phi(\ell)}$ by $\prod_{j\in\operatorname{im}\phi} z_j$. This proves the identity, including cancellations. Linearity handles cancellations between source monomials.

Terms of canonical degree below r have insufficient exterior degree; terms above r vanish. At degree r the coefficient of $e_{[r]}$ is the permanent of the displayed matrix, equal to its determinant in characteristic two. This is precisely the alternating polarization of the canonical degree-r component, and hence the top form under the stated promise. □

The operation groups factors belonging to one Boolean fiber before nilpotence enforces distinct fibers. Direct substitution into the unreduced affine product would instead discard the very collisions that must be retained. In the first control above, the two-factor recurrence contributes $w + w^2 = w$, canceled by the single-factor term. In the quartic-to-cubic control the repeated factors form one block, giving α, as required. If the actual degree exceeds r, equation (13) still computes its homogeneous component but need not equal the whole function's r-fold differences. Such histories remain in the failure budget.

Store exterior elements in 2^r-entry arrays. Multiplication by a degree-one vector obeys

$$(vF)_I = \sum_{i\in I} v_i F_{I\setminus\{i\}},$$

and costs $O(r2^r)$ coefficient operations. There are at most 3^D pairs (A, B) with $B \subseteq A$. Thus each source monomial costs

$$O(2^D[mr(D+1) + D + 1] + 3^D r 2^r) \tag{14}$$

primitive coefficient operations and $O(2^{D+r})$ coefficient entries of space. Construct all v_B explicitly; $r2^D \le 2^{D+r}$ absorbs their storage. Process monomials and contraction entries sequentially. A constant monomial has zero r-form. For the policy, $D, r \le d$, so the exponential work and storage are bounded by 6^d and 4^d, respectively. These costs are paid next.

6 Polynomial anchors and bounded descent

Sample each coordinate of each of the $r - 2$ anchors independently from

$$\mathcal{A}_k = \left\{ \sum_{j=0}^{k-1} a_j t^j : a_j \in \mathbb{F}_2 \right\} \subset \mathbb{F}_2(t).$$

This is a set of 2^k distinct field elements; no irreducible-polynomial search is needed for the recursive policy. Apply Lemma 7 over $\mathbb{F}_2[t]$, with the last two directions the binary coordinate vectors. At exterior mask I, every coefficient has t-degree at most

$$|I \cap [r-2]|(k-1).$$

Each new anchor slot contributes at most $k-1$, each binary slot contributes zero, and repeated slots vanish. Induction through the recurrence proves this bound even above the degree promise. Consequently every stored component coefficient and matrix entry has degree at most

$$D_r = (r-2)(k-1). \tag{15}$$

6.1 Actual rank and a polynomial kernel basis

Let $N \in \mathbb{F}_2[t]^{m\times m}$ have entry degree at most D. Full-pivot Schur elimination over $\mathbb{F}_2(t)$ computes its actual rank R_s and an attaining minor. Store reduced numerator/denominator pairs, reducing after every operation. After j pivots, each active entry is

$$\frac{\det \begin{bmatrix} P_j & c \\ b & a \end{bmatrix}}{\det P_j},$$

where P_j is the original selected nonsingular minor. This follows from the block determinant identity; a nonzero next pivot enlarges that minor. Numerator and denominator degrees are at most $(j+1)D$ and jD. Scalar-operation temporaries have degree at most a constant multiple of $1+mD$, and polynomial Euclid reduces them. Thus $O((m+1)^3(1+mD)^3)$ bit work is a conservative rank/minor cost. The product of pivots telescopes to the corresponding original determinant, so determinant computation has the same controlled degrees.

For $0 < R_s < m$, reorder selected rows and columns as $[P\ Q]$, and form

$$G = \begin{bmatrix} -\operatorname{adj}(P)Q \\ \det(P) I_\nu \end{bmatrix}, \qquad \nu = m - R_s. \tag{16}$$

Undo the column permutation to recover original coordinates. The columns lie in the entire rational kernel: the selected rows span the rational row space. They are independent because their bottom minor is $\det(P)^\nu \neq 0$. Every entry has degree at most R_sD. Compute the adjugate by sequential cofactor determinants; this elementary implementation has cost $O((m+1)^5(1+mD)^3)$. Use $G = I_m$ if $R_s = 0$, and the empty basis if $R_s = m$. No generic-rank oracle or enumeration of all maximal minors is used.

6.2 Repair at zero with a fixed valuation budget

Lemma 8. *Starting from* (16)*, at most $\nu R_s D$ binary column-combination and exact division steps produce a polynomial basis of the same rational kernel whose value at zero has rank ν. Entry degrees never increase. The computation takes $O((m+1)^5(1+mD)^3)$ bit work and $O((m+1)^2(1+mD))$ live bits, including rank and basis construction.*

Proof. While $G(0)$ has dependent columns, find nonzero $c \in \mathbb{F}_2^\nu$ with $G(0)c = 0$ and choose j with $c_j = 1$. Replace only column j by

$$G'_j = \frac{\sum_i c_i G_i}{t}. \tag{17}$$

The numerator has zero constant term in every coordinate, so division is an exact coefficient shift. The replacement remains in $\ker N$, since multiplication by t is cancellable. The column transformation has rational determinant $1/t$; independence and the rational span are preserved. Binary coefficients and division ensure no degree growth.

Fix the ν rows that initially gave the bottom minor in (16), and denote its determinant by $M(t)$. In each repair, determinant multilinearity gives $M' = M/t$: all terms except c_j duplicate another column and vanish. Thus the nonnegative valuation $v_t(M)$ drops by exactly one. Initially

$$v_t(M) = \nu v_t(\det P) \le \nu R_s D.$$

If $G(0)$ remains dependent then every maximal minor vanishes at zero, so the fixed minor still has positive valuation and another repair is possible. Termination follows within the stated budget; it may occur before this particular minor has valuation zero.

Binary dependence costs $O((m+1)^3)$ per step, and a polynomial column combination and shift cost $O((m+1)^2(1+mD))$. Multiplying by the repair budget and adding the rank/cofactor work gives the stated conservative bound. The fixed minor is an analysis potential, not a determinant recomputed by the implementation. Store reduced polynomial arrays and sequential cofactor workspaces; their degrees give the stated space bound. □

Set $V = G(0)$ and $U = \mathrm{im}_{\mathbb{F}_2} V$. Suppose the selected sample has full generic rank. Lemma 3, used with $E = \mathbb{F}_2(t)$, gives the strong identity on its rational kernel. This use specializes polynomial identities on a pivot chart with nonzero determinant in $\mathbb{F}_2(t)$; it is not a homomorphism of an arbitrary multivariate function field. Every repaired column still belongs to that kernel. Hence

$$T(e_i, G_{j_1}(t), \ldots, G_{j_{r-1}}(t)) = 0$$

is a polynomial identity with constant binary coefficients. Evaluation at zero preserves it, giving $T(\mathbb{F}_2^m, U^{r-1}) = 0$. Lemma 8 supplies $\dim U = m - R_s$, so the codimension is exactly R_s, without a factor k. The degree-r coefficients vanish on every affine coset by the top-difference characterization, giving degree at most $r-1$.

This argument makes no assertion that $N(0)$ retains rank or that $U = \ker N(0)$. In general that kernel is larger and fails the degree-reduction property. For arbitrary sampled matrices the repair and its costs remain valid; the strong-identity conclusion is used only under the good-history/full-generic-rank event. Zhang's valuation descent [1, Lemma 3.14] motivates the operation. One variable, bounded coefficient degrees and the explicit potential above supply the computational budget.

6.3 A query-preserving cutoff

Lemma 9. *Every nonaffine Boolean phase has Fourier spectral norm at least two.*

Proof. Choose two indices i, j in a nonzero monomial of degree at least two. The mixed Boolean derivative $D_i D_j p$ is nonzero: distinct monomials containing i, j give distinct residual monomials. Some binary assignment of the other variables makes it one, since a nonzero squarefree polynomial cannot vanish on the whole Boolean cube. Restriction at that assignment leaves $x_i x_j$ plus affine terms. Its phase has four Fourier coefficients of magnitude $1/2$ and norm two. Restriction groups Fourier coefficients with signs, so it cannot increase their ℓ_1 norm. Thus $B \ge 2$. □

For original $d \ge 3$ put $H_d = (2^{2d-1} - 8)/3$ and $Q_{\min} = 2 + H_d$. The spectral allowance in (1) is $Q_B = 1 + (1 + H_d)\log B \ge Q_{\min}$. Compute the threshold from n, d before any polynomial-matrix or

exponential table allocation. If $Q_{\min} \geq n$, read all n coordinate parities and evaluate the original p; this is exact and uses $n = \min(n, Q_B)$ queries. Otherwise

$$4^d < 6n + 4. \tag{18}$$

Only this latter branch invokes the recurrence and recursion. Threshold integers have $O(d)$ bits, even when their values exceed n. The test uses neither B nor a residual-degree oracle. Original $d \leq 2$ uses the exact leaf instead.

7 The recursive policy and its correctness

Use the parameters of Theorem 1 and apply the cutoff first. For a fixed history with degree at most r, choose a nonzero generic minor for analysis only. Its total anchor degree is at most $m(r-2) \leq nd$. The finite-set polynomial zero bound applies over $\mathbb{F}_2(t)$ to independent choices from $\mathcal{A}_k$, so one sample misses generic rank with probability at most $nd/2^k \leq 1/8$. This tests whether the determinant is the zero polynomial in t, not whether it vanishes at zero. Sharing the indeterminate does not couple the independently sampled coordinate values. Larger minors vanish identically before substitution, so no sampled rank exceeds the generic rank R; $R = 0$ succeeds surely.

Compute exactly t_A candidate matrices by Lemma 7, find their actual ranks over $\mathbb{F}_2(t)$ and keep the first maximum-rank candidate with its attaining minor. If any sample attains R, the selected one does too. Hence

$$\Pr[\text{selected rank} < R] \leq 8^{-t_A} \leq 2^{-t_A} \leq \epsilon/(2d). \tag{19}$$

This is amplification, not an executable certificate of generic rank. Apply the bounded descent only to the selected matrix. There is no unbounded retry, field-construction failure, sampled degree-drop test or exponential coset verification in the policy. The exact evaluator introduces no additional probability loss.

7.1 Lifting queries and updating the affine map

Binary elimination constructs L_C with $L_C C = I_m$. To ask for λz, ask the allowed original-input query $(\lambda L_C)x$ and add the known offset $(\lambda L_C)a$. From the repaired basis V, compute independent binary equations H with $\ker H = \operatorname{im} V$ and rank $\rho = R_s$. The answers determine $Hz = y$. Compute a solution z_0 and use the repaired basis V to replace

$$a \leftarrow a + Cz_0, \qquad C \leftarrow CV, \qquad m \leftarrow m - \rho. \tag{20}$$

All answer vectors are consistent because H has full row rank. The new columns remain independent and describe exactly the remaining input set. The map uses at most $n + n^2$ bits, independent of the number of previous batches. Even on a bad rank history, the true x remains in this set.

On a good stage, repaired specialization and the vanishing identity lower the degree on every coset to at most $r - 1$. The same original source and the updated map therefore supply the next form through Lemma 7. After two stages the map is $a + Cz_0 + CVz_1$, with linear part CVV_1 and degree at most $r - 2$. This proves arbitrary repeated composition without expanded ANFs or actual-degree detection.

7.2 Algorithm and conditional history

1. Normalize the original ANF, determine its degree d, and retain it throughout. If $d \le 2$, use the exact leaf routine below.
2. Compute $Q_{\min}$. If $Q_{\min} \ge n$, read all coordinates, evaluate p and return. Otherwise set $a = 0$, $C = I_n$ and $m = n$.
3. For scheduled $r = d, d-1, \dots, 3$: if $m = 0$, return $p(a)$. Otherwise compute t_A independent candidate matrices by (10)–(13); retain a maximum-rank matrix and attaining minor; construct and repair its polynomial kernel basis; compute independent binary equations; ask the lifted parities; and apply (20).
4. Apply the quadratic/affine leaf routine to the current source evaluator and return its bit.

Fix the hidden x. Condition on the complete history preceding the first failed rank stage. The current a, C, u are fixed, the degree promise follows inductively, and future anchor bits remain independent and uniform. Equation (19) applies to this conditional history. A first-failure decomposition over at most $d-2$ visited stages yields

$$\Pr[\text{a visited bad rank stage}] \le (d-2)\epsilon/(2d) \le \epsilon. \tag{21}$$

On its complement the final degree is at most two, so the leaf is exact. On other histories all loops, eliminations and affine updates still terminate. Only correctness, the propagated degree promise and the smaller query estimate may fail. This proof concerns every fixed input separately. It does not union-bound over an entire tree or supply an all-input correct random seed.

7.3 Exact quadratic and affine leaves

For $m = 0$ return $p(a)$. Otherwise evaluate u at zero, the e_i and the $e_i + e_j$ for $i < j$, and set

$$c = u(0), \quad l_i = u(e_i) + c, \quad b_{ij} = u(e_i + e_j) + u(e_i) + u(e_j) + c.$$

The explicit quadratic $g(z) = c + \sum_i l_i z_i + \sum_{i<j} b_{ij} z_i z_j$ equals u on a good history. On a bad history it is simply a defined quadratic approximation. Form its alternating polar matrix B_g. Symplectic elimination constructs invertible P with $P^{\mathsf{T}} B_g P$ consisting of t alternating pairs and a zero block, where $\operatorname{rank} B_g = 2t$. Explicitly choose a pair a_0, b_0 with pairing one, and replace each remaining vector v by

$$v + B_g(v, b_0)a_0 + B_g(v, a_0)b_0.$$

It is then perpendicular to that pair. Repeating preserves a basis and isolates the radical; explicit bookkeeping costs at most $O(m^4)$ bit work. The canonical Boolean expression is

$$g(Pw) = c + \sum_{i=1}^{t} w_{2i-1} w_{2i} + \ell \cdot w, \qquad \ell_j = g(Pe_j) + c.$$

Computing these linear terms includes squares reducing to linear terms. Query the t odd coordinates, lifting through CP. Once they are known, the remaining expression is affine: at most one additional parity gives its value. The leaf uses at most $t+1$ queries and at most m; if $t = 0$ it uses zero queries for a constant and at most one otherwise.

On a good history, $\|\widehat{(-1)^u}\|_1 = 2^t$. Indeed, each independent two-bit quadratic pair has four Fourier coefficients of magnitude $1/2$ and norm two; affine characters shift frequencies and invertible linear changes permute them. Affine restriction cannot increase spectral norm: it groups original Fourier coefficients with signs, so the triangle inequality applies. Thus the leaf costs at most $1+\log B$ with B from the original input.

7.4 Total queries

On the good event, stage r uses at most $2^{2r-3}\log B$ queries, including zero when the actual degree is smaller than r. Summing with the leaf gives

$$1+\left[1+\sum_{r=3}^{d}2^{2r-3}\right]\log B = 1+\frac{2^{2d-1}-5}{3}\log B.$$

Every batch removes exactly as many dimensions as queries asked, and the leaf uses at most the remaining dimension. Hence the cap n holds on all tapes. Taking the minimum and using $B\le\sqrt{s}$ proves (1). The cutoff branch already satisfies the same bound exactly. The additive one covers a nonconstant parity with $s=1$.

8 Total recursive computation and space

Let S_0 be the original sparse encoding length, $L\le S_0+1$ its normalized monomial count, and put

$$P_0=n+S_0+b+2,\quad \Delta=d(k-1),\quad \Lambda=1+n\Delta,\quad V_0=n^2+(d+1)(S_0+n)+1.$$

Normalization by repeated scans is polynomial; V_0 bounds a source evaluation with its affine transformation. From (14), a per-entry coefficient-operation allowance is

$$E_0=O(S_0+(L+1)[2^d(nd(d+1)+d+1)+d6^d]).$$

Schoolbook binary-polynomial arithmetic of degree at most Δ costs $O((\Delta+1)^2)$ per coefficient operation. Include every sampled matrix, normalized rational elimination, selected adjugate and repair, binary equations, query lifting, affine update and exact leaf. With indexed bit storage, a conservative total is

$$\begin{aligned}T_0=O(P_0^4+dt_A[n^2(\Delta+1)^2E_0+(n+1)^3\Lambda^3]\\ +d(n+1)^5\Lambda^3+d(n+1)^3+n^2V_0+n^4).\end{aligned}\tag{22}$$

The first term pays normalization, threshold/precision arithmetic and reading. Query descriptions total at most n^2 bits and answers at most n bits; constructing them is charged here while oracle calls are counted separately.

Stream source monomials, contraction entries and candidates; retain one candidate, one best matrix, bounded rational/cofactor workspaces, one polynomial basis, affine maps and counters. A live bit-space bound is

$$W=O(S_0+(n+1)^2\Lambda+(\Delta+1)4^d+ndk+n^2+b+d+1).\tag{23}$$

Cofactor determinants are sequential. No full residual ANF, decision tree or random tape is stored. The random-bit cap is

$$t_Akn\sum_{r=3}^{d}(r-2)=t_Akn(d-2)(d-1)/2,$$

with zero random bits in the read-all and original low-degree branches.

The cutoff gives $4^d=O(P_0)$ and $6^d\le 16^d=O(P_0^2)$ before these arrays are allocated. Also $n,d,L,k,t_A=O(P_0)$, $\Delta=O(P_0^2)$, $\Lambda=O(P_0^3)$ and $E_0=O(P_0^5)$. Thus (22)–(23) imply $T_0=O(P_0^{15})$ and $W=O(P_0^5)$. Sequential scans can simulate indexed access with conservative overhead

$W^2(T_0+1)$; $O(P_0^{28})$ is a loose common sequential bit-time bound including controls and endpoint branches. These exponents are not claimed optimal or practical.

On bad histories the canonical component evaluator remains defined with the same coefficient-degree bound. Descent accepts every bounded-degree matrix; all loops have the same caps and the original cutoff is unchanged. Time, space and termination therefore hold on every tape. Separately encoded arbitrary rational precision of length A adds its own polynomial parsing cost. This completes Theorem 1. There is no variance, normalization or approximation loss; only rank sampling contributes error.

8.1 Accounting for the archived implementation

The complete implementation retains the original source as bit masks and caches factor subsets and anchor components. It does not satisfy the exact streamed expression (23) without an extra charge. In addition, allow

$$O(nL + L2^d(n+dk+1) + n^3 + dt_A \log(n+d+k+b+2)) \tag{24}$$

bits for mask representation, caches, a mask-list quadratic leaf and retained path summaries. Query transcripts use $O(n^2)$ bits. The cutoff absorbs (24) into $O(P_0^4)$, hence the common $O(P_0^5)$ space bound. Bit-mask source evaluations cost $O(n^2+nL)$, giving $O(n^4+n^3L)$ for the leaf; normalization can be charged $O(nL^2)$ without average-case hashing assumptions. Cache preparation and these extra costs fit the common polynomial time bound.

The read-only reference observer receives intermediate states and returns no information used by the policy. Its truth tables, explicit residual ANFs and determinant enumeration are separately bounded test work and are not included in the general polynomial policy claim. An executable implementation and finite tests do not constitute a machine-checked refinement proof.

9 Controls and limitations

Write $x_I = \prod_{i\in I} x_i$. Four exact controls are:

Polynomial	Anchors	Generic rank	Binary kernel
x_{123}	e_1	2	$\langle e_1\rangle$
$x_{123}+x_{456}$	e_1+e_4	4	$\langle e_1, e_4\rangle$
$x_{123}+x_{145}$	e_1	4	$\langle e_1\rangle$
$x_{1234}+x_{1256}$	e_1, e_2	4	$\langle e_1, e_2\rangle$

The contractions attain these ranks by nonsingular alternating pairs. Matching generic upper bounds follow from alternating rank in dimensions three and five, the two independent three-dimensional blocks in the second example, and the two independent generic anchors in the radical in the quartic example. The listed kernels lower degree on every coset. Adding terms of degree at most $d-1$ leaves T unchanged and preserves the target. In the disjoint-cubic example, contraction only at e_1 has rank two and leaves the second cubic intact: nonzero sampled rank is insufficient.

The field-to-binary loss is real. In the first cubic, take an anchor $(1,\alpha,0)$ with $\alpha \in E \setminus \mathbb{F}_2$. The extension-field rank is two and the kernel is $E(1,\alpha,0)$, but its binary intersection is zero. Binary codimension is three, not two. This does not establish sharpness of the factor k. Zero anchors give rank zero and can fail degree reduction; these are covered by the exceptional event.

The estimate is nonvacuous when $4^d \log s = o(n)$. This regime need not arise by adjoining unused variables: the product of three parities on disjoint growing variable blocks depends on

every variable, has degree three, sparsity eight and at most n^3 ANF monomials. The input cost remains paid.

For an explicit two-stage control take

$$p = x_1x_2(x_3x_4 + x_5x_6) + x_7x_8x_9.$$

Quartic anchors e_1, e_2 give rank four, with pairs $(3, 4)$ and $(5, 6)$. This is generic rank: the two generic anchors lie in the radical of the six-dimensional support, and the last three coordinates are inactive. Querying x_3, x_4, x_5, x_6 leaves

$$u(z) = cz_1z_2 + z_3z_4z_5, \qquad c = x_3x_4 + x_5x_6.$$

The cubic contraction at e_3 has generic rank two, since its alternating three-dimensional block has rank at most two. Querying z_4, z_5, equivalently x_8, x_9, leaves $cw_1w_2 + (x_8x_9)w_3$, of degree at most two for every answer branch. This control illustrates the general composition proof; finite agreement is not used as its substitute.

The recursive construction stores a compact affine map rather than a residual ANF. Its subset work is used only after the cutoff makes that work and storage polynomial. The policy now matches the retained existential query scale, while construction of a globally correct deterministic tree remains a different obligation. Reading all n hidden bits already permits direct evaluation of the explicit p. Our result concerns fewer parity queries with paid computation, not faster unrestricted evaluation or a general quantum simulator.

Earlier parity-tree recursion is due to Tsang–Wong–Xie–Zhang [3]. Zhang's quantum protocol [4] assumes an efficiently encoded/decoded Fourier spectrum, a different access model and output. Hrubeš's multilinearization hardness result [5], Theorem 7(ii), concerns uniformly small arithmetic representations of whole multilinearizations. It does not rule out our exponential-in-degree component evaluator combined with the query-preserving cutoff; no compact representation of the whole multilinearization is produced. Determinantal sieving [6] also uses characteristic-two algebra with field evaluation access, but multilinear or odd support differs from Boolean reduction: z^2 reduces to z on the Boolean cube. The factor-partition identity supplies that missing interface before exterior evaluation. These comparisons explain the input distinction; they do not establish novelty. Unrestricted XOR log-rank, stronger AA bounds and BPP versus BQP are not resolved.

9.1 Complete-policy validation and rank-collapse controls

The archive includes a complete implementation and separate exact reference. An initial full-policy campaign completed 21 trajectories: 19 independently certified good-event paths returned the correct result within the spectral cap; two deliberately bad coefficient tapes returned incorrect results within the unconditional caps, as allowed by the Monte Carlo contract. It exercised repeated reductions, non-coordinate queries, a nonzero affine offset and nonvacuous kernel identities. Its final aggregate coverage gate failed because no selected matrix lost rank at zero. The 3,868 basis repairs did not imply that distinct matrix-rank event. This campaign remains recorded as STOP, preceded by a preserved startup import failure and its single permitted correction.

A separately preregistered check then covered total and partial rank collapse using the unchanged policy. For the first fixture, index coordinates from zero and put

$$A = x_0 + \cdots + x_3, \quad B_1 = x_4 + \cdots + x_7, \quad C_1 = x_8 + \cdots + x_{12}, \quad p = AB_1C_1.$$

The anchor $t(e_0 + e_4)$ gives

$$N(t) = t[(A + B_1) \otimes C_1 + C_1 \otimes (A + B_1)].$$

Its generic rank is two and $N(0)$ has rank zero. The repaired kernel is $\ker(A+B_1, C_1)$, of dimension 11. On a coset $A + B_1 = a, C_1 = c$, Boolean reduction gives $p = c(1 + a)A$, of degree at most one. The unrepaired basis has rank zero, whereas the whole kernel of $N(0)$ retains the original cubic; both shortcuts fail their required contracts.

For the second fixture let $n = 128$ and

$$p = x_0x_1x_2x_3 + x_4x_5x_6x_7, \qquad h_1 = e_0 + e_4, \quad h_2 = e_1 + te_5.$$

The only nonzero contraction entries are $N_{23} = N_{32} = 1$ and $N_{67} = N_{76} = t$. Generic rank is four (each disjoint volume contraction has rank at most two), but rank at zero is two. The entries have polynomial content one, so this is not solely a common scalar factor. Rational-kernel descent fixes coordinates 2,3,6,7 and returns dimension 124; the entire kernel at zero has dimension 126 and leaves the second quartic intact. Every valid answer coset instead has degree at most two.

Fixture	Matrix ranks	Repaired dimension	Repairs	Queries
Full-support cubic	$2 \to 0$	11	22	3
Disjoint quartics	$4 \to 2$	124	248	7

Both complete trajectories passed: outputs 1 and 0 at hidden bit masks 273 and 255. Their original spectral norms are 5/2 and 121/16, with exact comparisons proving $Q_B < n$ in each case. All 6,523 assertions passed, including 15 sampled matrices, 21 answer cosets and 21 coordinate-derivative restrictions. Four negative-control predicates reject the unrepaired basis and naive whole kernel at zero. One bounded run used 14.196 seconds and 22,769,664 bytes observed peak working set under 120 seconds and 128 MiB. These measurements describe the retained run, not asymptotic complexity or a random success-frequency estimate. Quartic padding serves only to enter the informative branch. The earlier STOP remains unchanged; this separate check closes its specific missing coverage.

Disclosure and status. OpenAI Codex assisted derivation, source comparison, implementation, drafting and document preparation. Review across mathematical, source, claims and artifact lenses is internal and AI-assisted. The archive preserves the earlier 256-fixture component diagnostic and the complete-policy evidence with failures, negative controls and provenance. No research experiment was rerun for this publication. General statements follow from the analytic proof and attributed structural imports, not finite agreement. No independent peer review or formal certification is claimed; novelty remains unestablished. PaperMill's queued certification covers the full revised theorem and its dependencies, including bit costs and the distinction between mathematical and executable algorithms.

References

[1] Shengyu Zhang. *Parity Decision Trees of Depth $O(4^d \log \|\widehat{f}\|_1)$ for Boolean Functions f of $\mathbb{F}_2$-Degree d.* Preprint, November 2, 2026. arXiv:2611.03452v1.

[2] Guy Moshkovitz and Daniel G. Zhu. *Schur complements for tensors and multilinear commutative rank.* Preprint, March 11, 2026. arXiv:2603.11038v1.

[3] Hing Yin Tsang, Chung Hoi Wong, Ning Xie and Shengyu Zhang. *Fourier sparsity, spectral norm, and the Log-rank conjecture.* Proceedings of FOCS, 2013. arXiv:1304.1245.

[4] Shengyu Zhang. *Efficient quantum protocols for XOR functions.* Proceedings of SODA, 2014. arXiv:1307.6738.

[5] Pavel Hrubeš. *On hardness of multilinearization, and VNP-completeness in characteristics two.* Author manuscript, April 21, 2015, Theorem 7(ii). Author-hosted manuscript.

[6] Eduard Eiben, Tomohiro Koana and Magnus Wahlström. *Determinantal Sieving.* TheoretiCS, volume 4, October 2, 2025. doi:10.46298/theoretics.25.21.